

WT Unique Questions

Module 1: Fundamentals of Wireless Networks & Spread Spectrum

#	Question	Appeared In
1	Explain the difference between an Infrastructure-based Network and an Adhoc Network of WLAN.	WT DEC 2023, WT MAY 2024, WT MAY 2025, WT DEC 2024
2	Compare Infrastructure-based Network and Adhoc Network.	WT MAY 2024
3	Explain Direct Sequence Spread Spectrum (DSSS).	WT DEC 2023
4	What is Frequency Hopping Spread Spectrum (FHSS), and how does it improve communication security?	WT DEC 2024
5	Compare Frequency Hopping Spread Spectrum (FHSS) and Direct Sequence Spread Spectrum (DSSS).	WT MAY 2025
6	Explain spread spectrum and briefly outline DSSS and FHSS.	WT MAY 2024
7	Compare FDMA, TDMA and CDMA.	WT DEC 2023
8	Explain how LoRaWAN enables long-range, low-power communication using Chirp Spread Spectrum.	WT DEC 2024
9	How does OFDMA handle synchronization and interference between users?	WT DEC 2024
10	Draw and explain OFDMA with a neat diagram.	WT MAY 2024

Module 2: Cellular Networks (GSM, GPRS, UMTS, 4G, 5G)

#	Question	Appeared In
1	Explain in detail GSM Architecture with a neat diagram.	WT DEC 2023

#	Question	Appeared In
2	Illustrate and explain GSM architecture with its interfaces. Compare GSM and UMTS.	WT MAY 2024
3	Explain the function of a Subscriber Identity Module (SIM) in GSM.	WT DEC 2024
4	What are the primary security mechanisms employed in GSM to protect user data and privacy?	WT DEC 2024
5	Explain UMTS and GSM security.	WT DEC 2023, WT MAY 2024
6	List and explain the core components of the UMTS network.	WT DEC 2024
7	Draw and explain UMTS network architecture and compare GSM and UMTS.	WT MAY 2023
8	Explain role of SGSN and GGSN in GPRS.	WT MAY 2025
9	Draw a neat diagram of GPRS system architecture and explain the function of each block.	WT MAY 2023
10	Draw and explain 4G architecture with its specifications.	WT DEC 2023
11	Explain the architecture and working of 4G.	WT DEC 2024
12	Draw and explain 4G network architecture and compare with 3G.	WT MAY 2025
13	Explain 4G network architecture with its specifications.	WT MAY 2023
14	What do you mean by massive MIMO?	WT DEC 2023, WT MAY 2024, WT MAY 2025
15	Compare 1G to 5G mobile standards (evolution/technological differences).	WT DEC 2023, WT MAY 2023, WT MAY 2024, WT MAY 2025
16	Write a short note on CDMA2000.	WT MAY 2023
17	Write a note on LTE frame structure in detail.	WT MAY 2023

Module 3: Wireless LAN (Wi-Fi / IEEE 802.11) & Cisco UWN

#	Question	Appeared In
1	Draw and explain IEEE 802.11 Architecture.	WT DEC 2023

#	Question	Appeared In
2	Draw and explain protocol architecture of IEEE 802.11. Also discuss power management.	WT MAY 2023
3	Draw and explain Wi-Fi Architecture and Protocol Architecture.	WT MAY 2024
4	Describe the system architecture of 802.11.	WT MAY 2025
5	How does Wi-Fi handle multiple users and avoid collisions in the network?	WT DEC 2024
6	Outline the method that supports mobility in CISCO Unified Wireless Network.	WT DEC 2023, WT MAY 2023, WT MAY 2024, WT MAY 2025
7	Draw and explain Cisco UWN lightweight AP and WLC Operation.	WT DEC 2023
8	Draw and explain the architecture of Cisco UWN with its features.	WT MAY 2023
9	Give the features of Cisco Unified Wireless Network and explain its architecture.	WT MAY 2025
10	How does Cisco implement security in its Unified Wireless Network?	WT DEC 2024
11	Illustrate the RF site survey process and its importance in designing a wireless network with Lightweight AP and WLC.	WT MAY 2024
12	Compare IEEE 802.11 and IEEE 802.16.	WT MAY 2025

Module 4: Wireless Security (WEP, WPA, WPA2)

#	Question	Appeared In
1	Give the significance of WEP protocols. What are the features of WPA2?	WT DEC 2023, WT MAY 2023, WT MAY 2025
2	Explain WEP and the security standard used.	WT MAY 2024
3	Discuss the flaws in WEP's initialization vector (IV) implementation and how it leads to key reuse and susceptibility to attacks.	WT DEC 2024

Module 5: Wireless Sensor Networks (WSN), ZigBee, Bluetooth, LoRaWAN

#	Question	Appeared In
1	What are the various challenges in WSN and explain the architecture of WSN.	WT DEC 2023, WT MAY 2025
2	Explain the layered architecture of WSN protocol and discuss issues and challenges in WSN.	WT MAY 2023
3	Outline the WSN architecture along with neat diagrams. State the various issues in WSN.	WT MAY 2024
4	Explain Zigbee protocol stack.	WT DEC 2023, WT MAY 2025
5	What is Zigbee? Explain ZigBee protocol stack in detail.	WT MAY 2023
6	State different features of Zigbee and explain its protocol stack.	WT MAY 2024
7	What is mesh networking in ZigBee, and how does it enhance network reliability?	WT DEC 2024
8	Draw and explain Bluetooth Architecture.	WT MAY 2025
9	Describe Bluetooth architecture and protocol stack. Also discuss its limitations.	WT MAY 2023
10	Explain the pairing and security mechanisms used in Bluetooth communication.	WT DEC 2024
11	Explain LoRaWAN.	WT MAY 2024
12	Draw and explain LoRaWAN network architecture and technology stack in detail.	WT MAY 2023
13	State the features of LoRaWAN.	WT MAY 2025

Module 6: WiMAX, VANET, MANET & Other Networks

#	Question	Appeared In
1	State the features of Wi-MAX.	WT DEC 2023, WT MAY 2024
2	Explain the architecture and features of a WiMAX network.	WT DEC 2024

#	Question	Appeared In
3	Draw and explain the architecture of Wi-MAX.	WT MAY 2024
4	Give the Features of VANET and MANET.	WT MAY 2024
5	Give the Features of VANET, E-VANET and MANET.	WT DEC 2023
6	Compare VANET and MANET.	WT MAY 2025
7	What role does VANET play in intelligent transportation systems (ITS)?	WT DEC 2024
8	Explain any four features of MANET and compare MANET and WSN.	WT MAY 2023